

MACHINE LEARNING IN PHYSICS FOUNDATIONS 1

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PHY6937 / PHY4936

Goals of this Course

1. Gain a good understanding of the mathematical basis of machine learning (ML), which is a foundational component of artificial intelligence (AI).
 2. Gain experience building ML models using **PyTorch** to solve data science problems in physics.
 3. Gain experience with different ML models.
 4. Gain an appreciation of the power of ML models as well as their (current) limitations.
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Course Evaluation

In addition to your **Course Participation** (attending, asking questions, contributing to discussions), your course grade will be based on:

1. **In-class Assignments**
 2. **Project with a write-up and a brief presentation**
 1. Graduates: A project related to your research area.
 2. Undergraduates: A project selected from a few options.
 3. Presentations: Last week of the semester.
 3. **Grade**: Graduates (S/U), Undergraduates (Letter Grade).
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What is Artificial Intelligence?

Artificial Intelligence

Algorithms that cause machines to exhibit *human-* or *superhuman-*level intelligence.

ChatGPT



Developer(s) OpenAI

Initial release November 30, 2022

Stable release August 7, 2025

Engine GPT-5

Large Language Model (LLM)

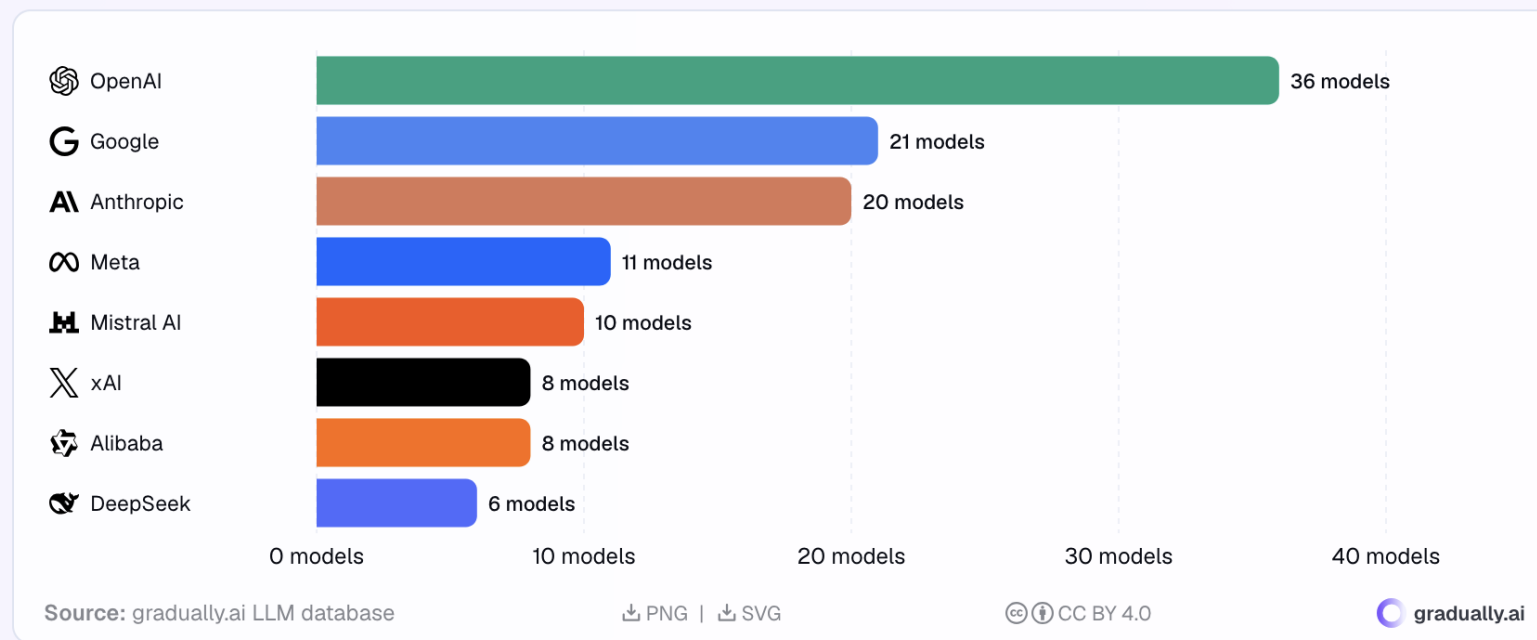


BY ANTHROPIC

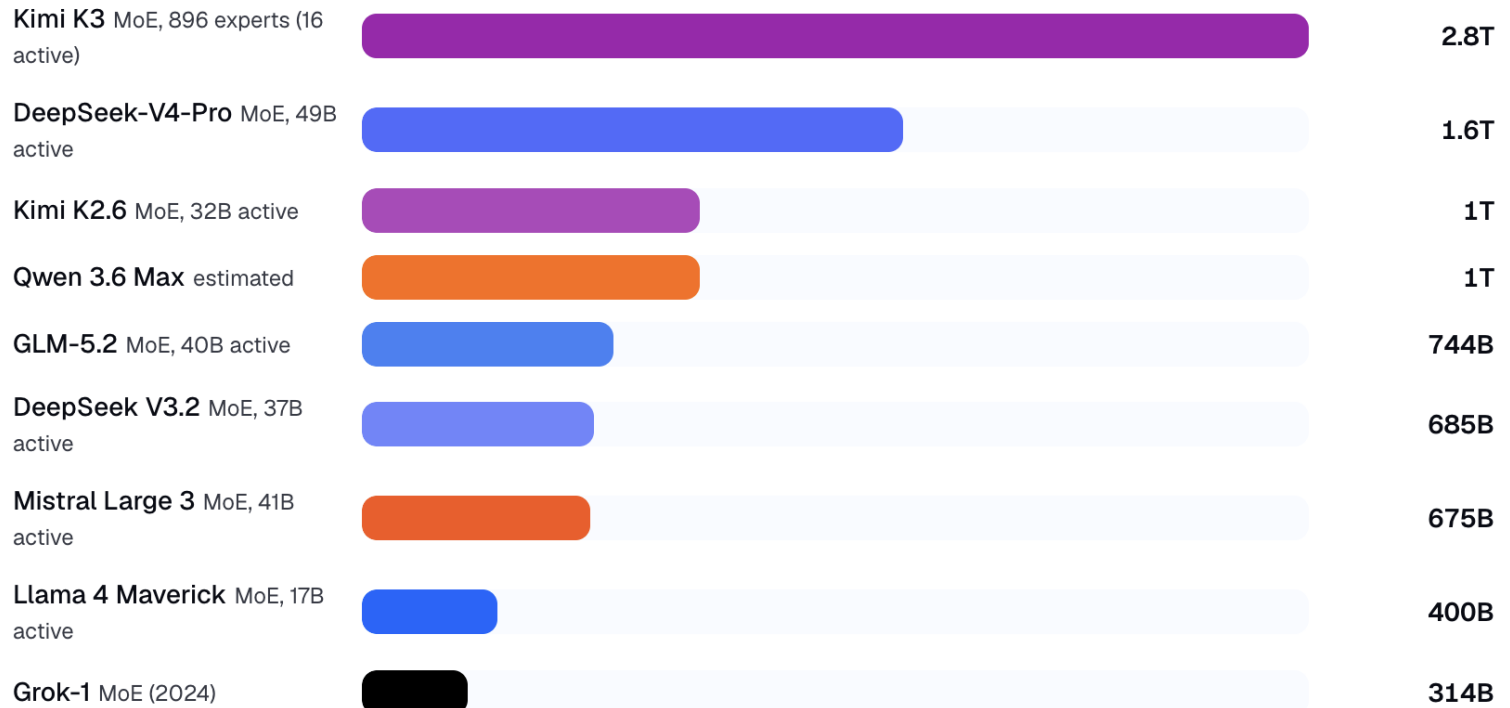
Large Language Model (LLM)

2. The Biggest LLM Providers by Model Count

A simple indicator of how active a lab is: the number of models it maintains. The chart below shows how many of the models we track belong to each provider:



<https://www.gradually.ai/en/llm-statistics/>



Source: gradually.ai LLM database

📄 PNG | 📄 SVG

CC BY 4.0

 gradually.ai

<https://www.gradually.ai/en/llm-statistics/>

Claude Demo

What is Machine Learning?

Artificial Intelligence

Algorithms that cause machines to exhibit human- or *super-human* level intelligence.

Machine Learning

Algorithms for modeling data.

What is Machine Learning?

Supervised Learning

Data: (x, y)

y are labels

Task: $x \rightarrow y$

Use cases:

- Classification, regression, translation, etc.

Unsupervised Learning

Data: x

no labels

Task: find structure in,
and/or model, data

Use cases:

- Clustering, data compression, solving differential equations, etc.

What is Machine Learning?

Generative Learning

Data: x

may or may not be associated with labels

Task: $x \rightarrow p(x) \rightarrow x$

Use cases:

- fast simulators, image/text generation, chatbots, etc.

Reinforcement Learning

Data: (x, a, r)

x state of the environment

a action taken on environment

r reward arising from action

Task: find optimal $x \rightarrow a$

Use cases:

- Game playing, robotics, accelerator controls, etc.

What is Deep Learning?

Artificial Intelligence

Algorithms that cause machines to exhibit human- or *super-human*-level intelligence.

Machine Learning

Algorithms for modeling data

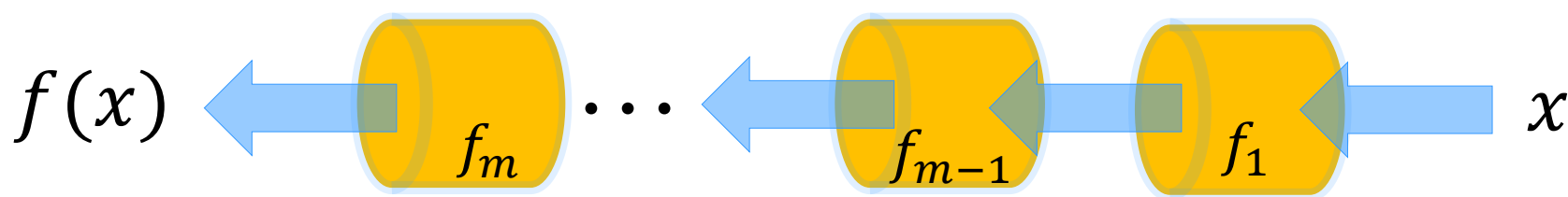
Deep Learning

ML using (large) neural networks

What is Deep Learning?

Deep learning is the science and art of fitting models to data using functions formed through *composition*:

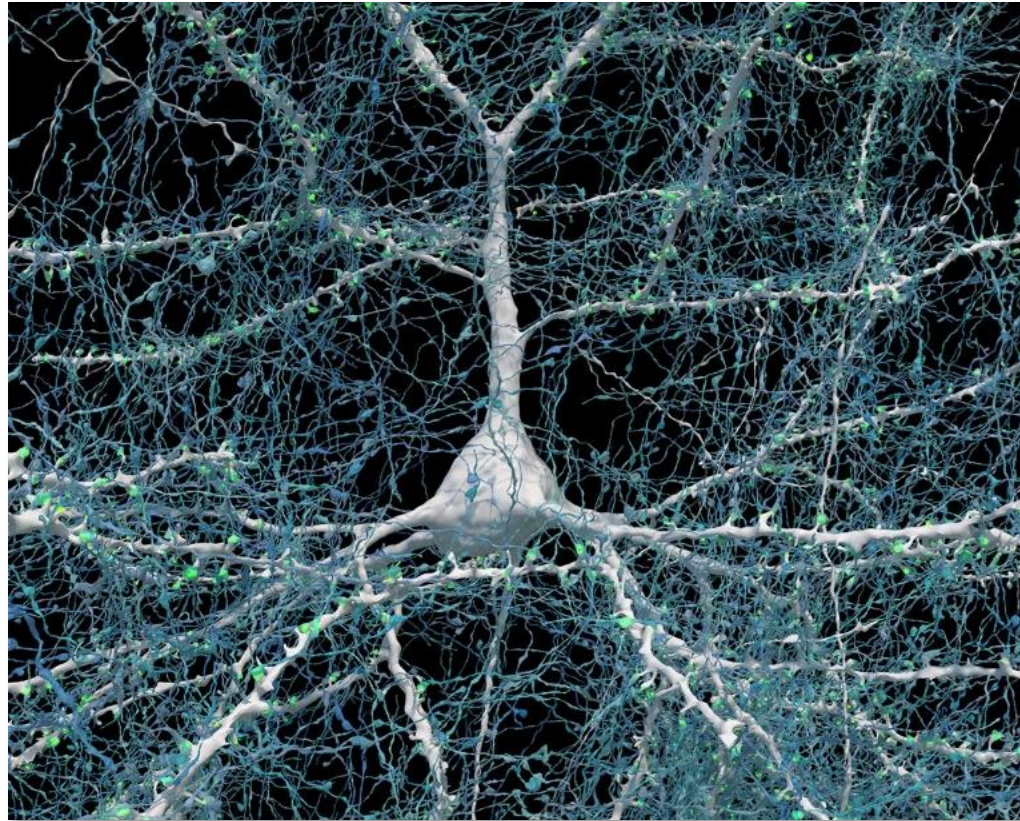
$$\begin{aligned} f(x) &= f_m \circ f_{m-1} \circ \dots \circ f_1 \\ &= f_m(f_{m-1}(\dots f_1(x) \dots)) \end{aligned}$$



Each of these functions is referred to as a **layer**. The ChatGPT3 LLM has **96** layers and **175 billion** parameters!

BASIC BUILDING BLOCK: THE PERCEPTRON

The Brain's Computational Unit



<https://www.nature.com/articles/d41586-024-01387-9>

Basic AI Comp. Unit: The Perceptron

$$y = g(xA^T + b)$$

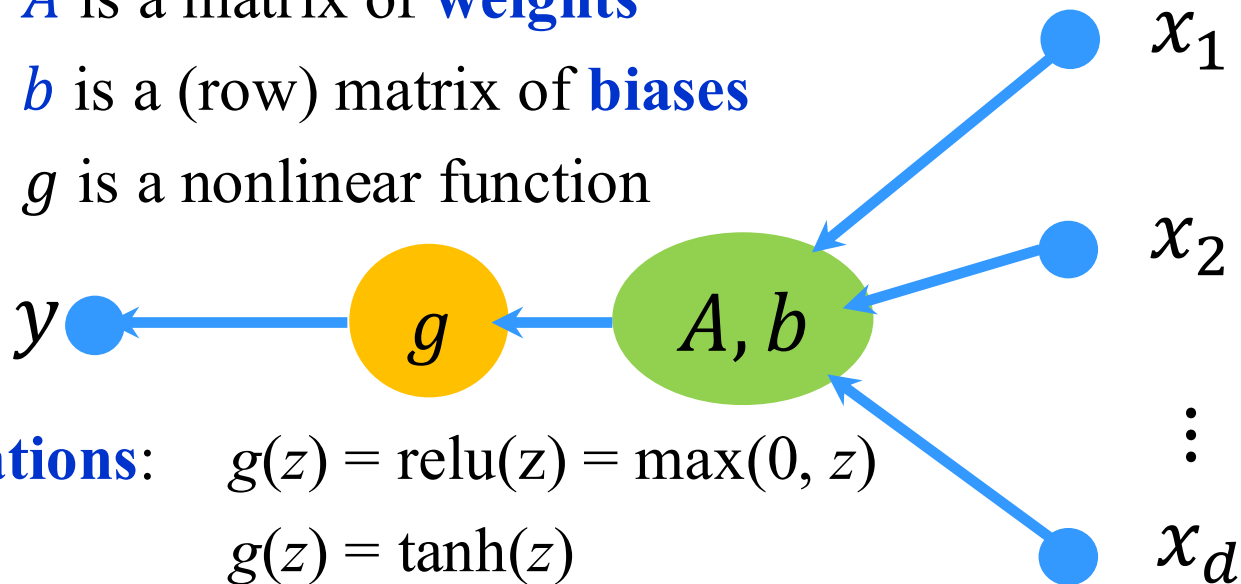
(Frank Rosenblatt, 1958)

x is a (row) matrix of input data

A is a matrix of **weights**

b is a (row) matrix of **biases**

g is a nonlinear function



Activations: $g(z) = \text{relu}(z) = \max(0, z)$

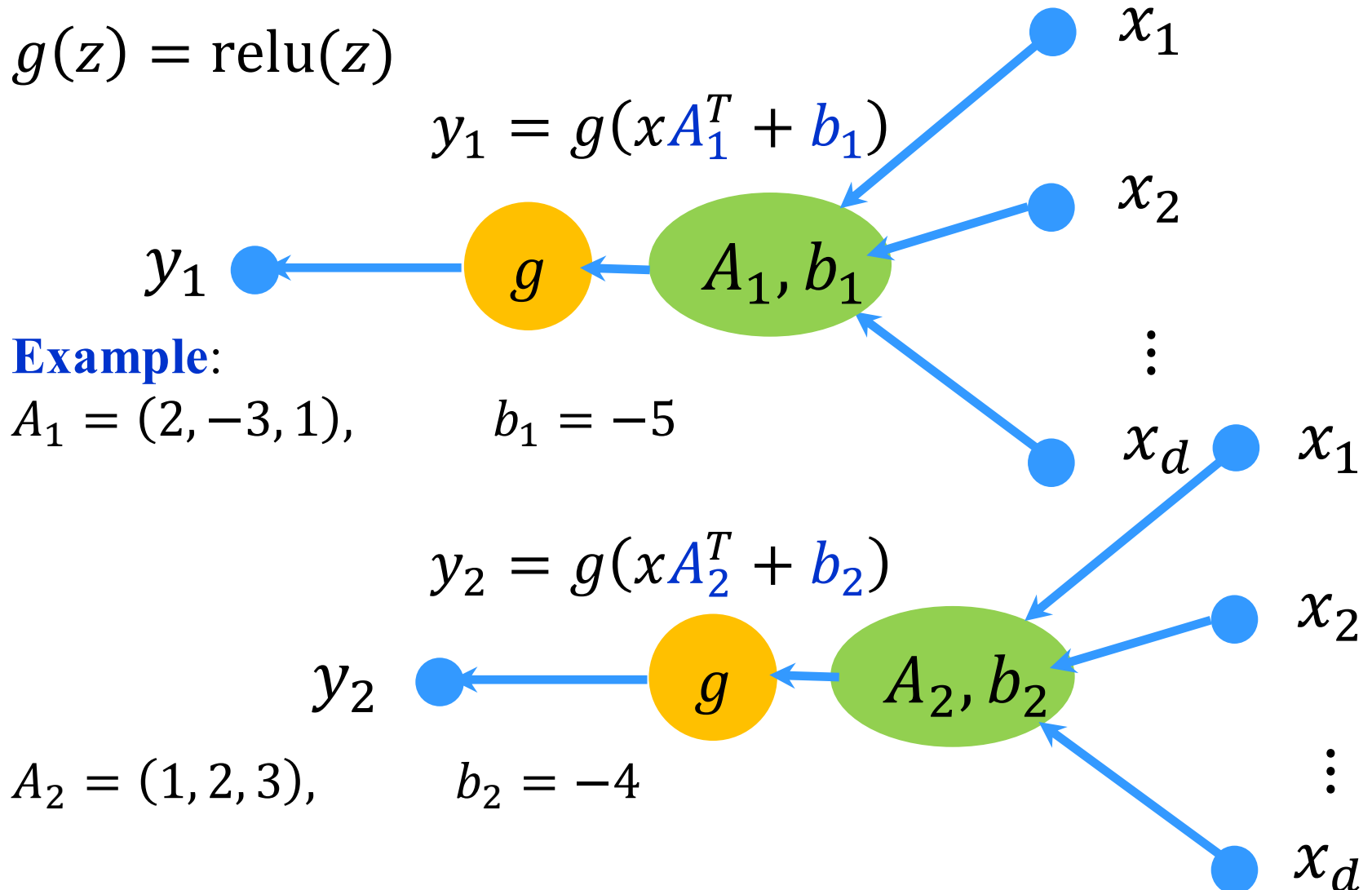
$g(z) = \tanh(z)$

$g(z) = \text{sigmoid}(z) = 1 / (1 + \exp(-z))$

<https://news.cornell.edu/stories/2019/09/professors-perceptron-paved-way-ai-60-years-too-soon>

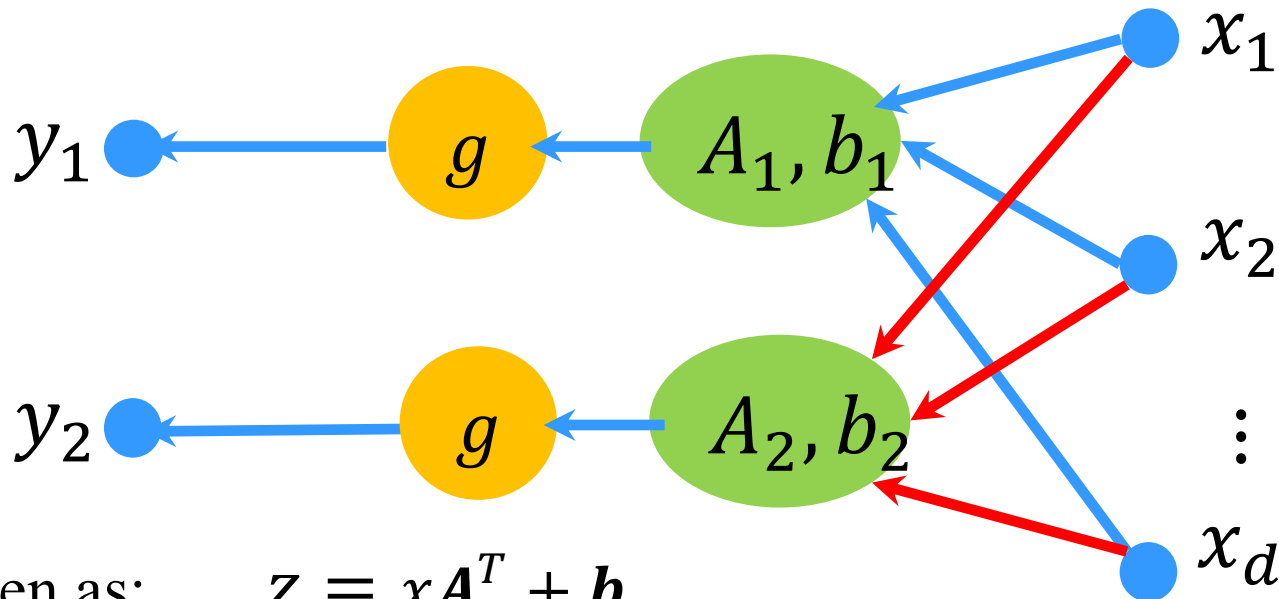
MULTI-NODE PERCEPTRON

Multi-Node Perceptron



Multi-Node Perceptron

A multi-node perceptron is often drawn as follows,

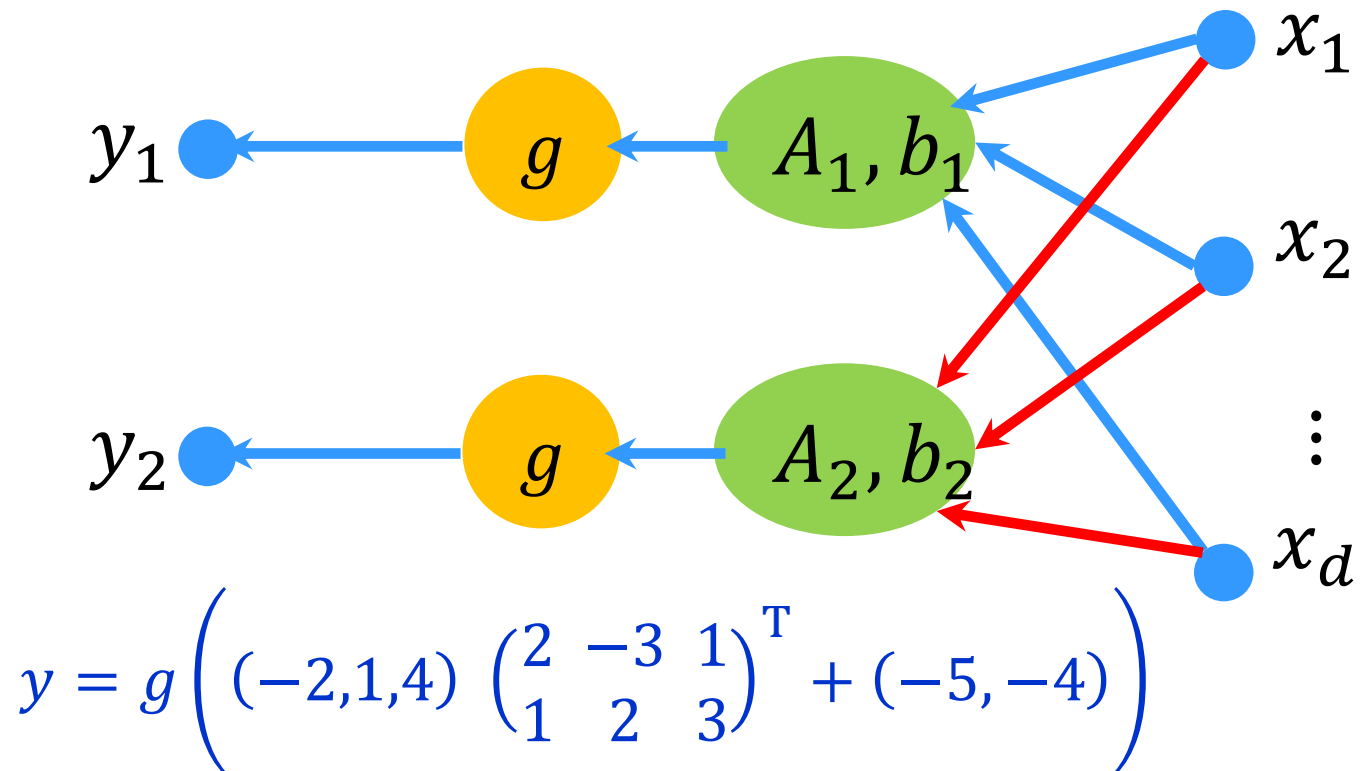


Written as: $z = xA^T + b$

with $y = g(z)$, usually, the applied *elementwise*,
that is, to every element of its matrix input.

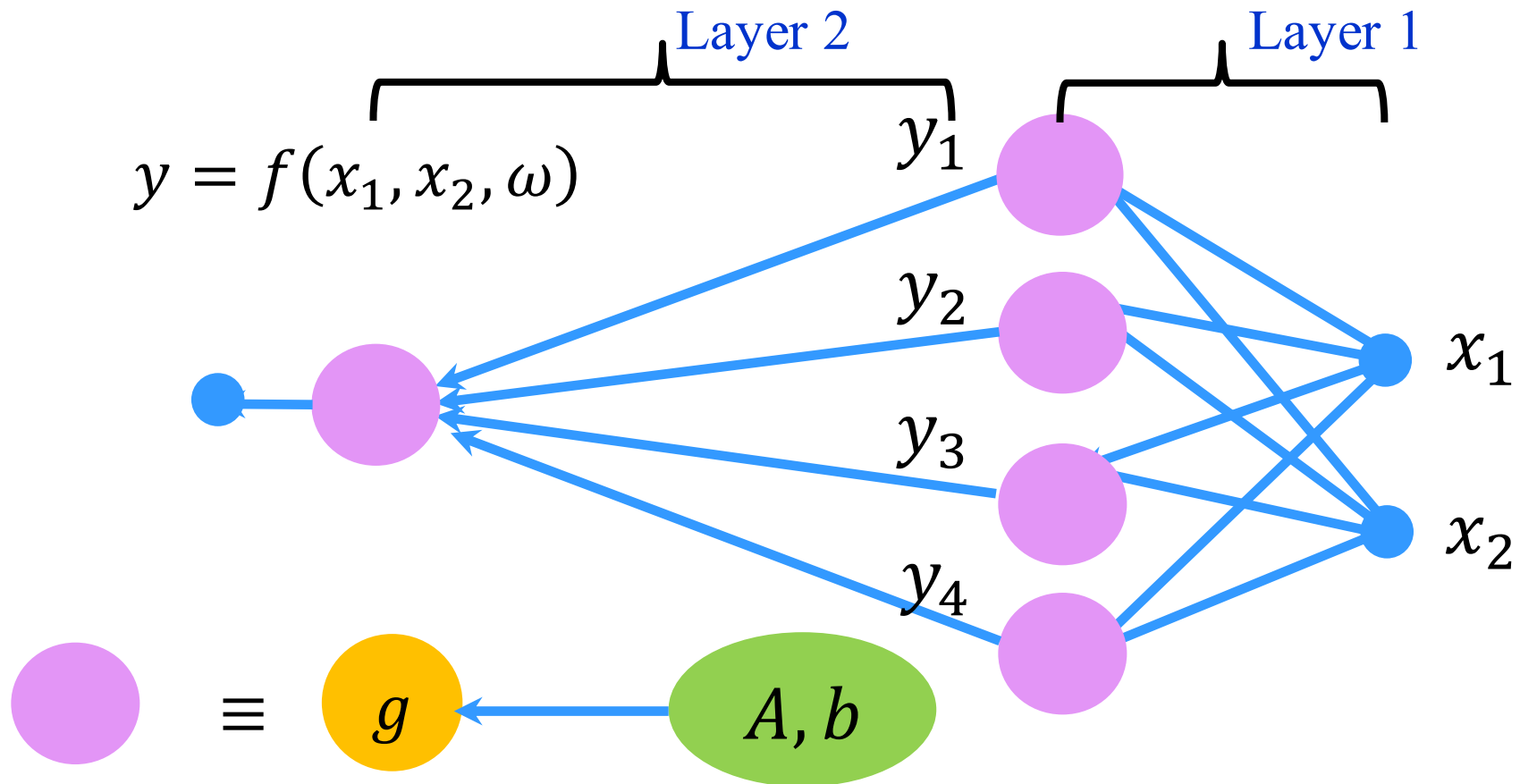
Multi-Node Perceptron: Example

$$x = (-2, 1, 4), \quad A = \begin{pmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \end{pmatrix}, \quad b = (-5, -4)$$



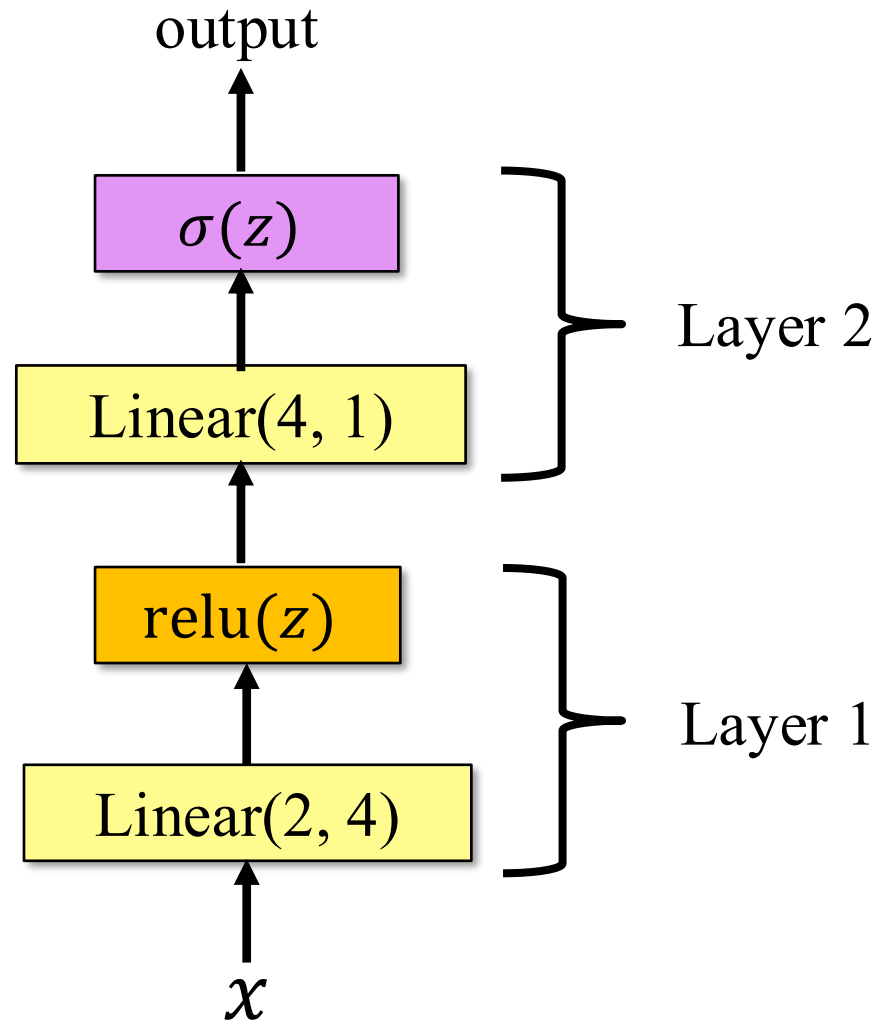
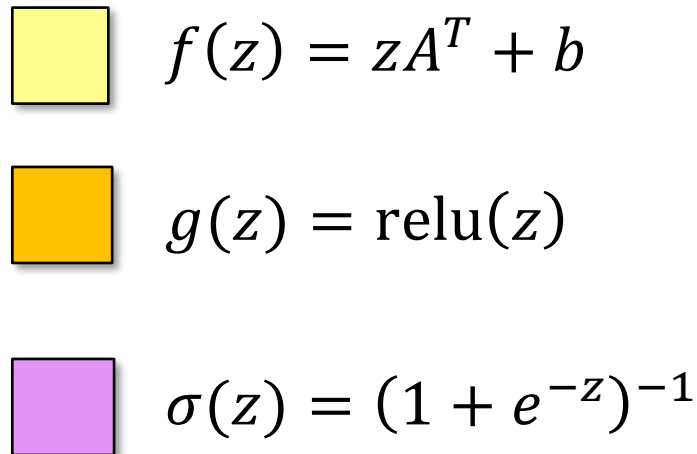
Multi-Node Perceptron

Consider modeling the data triplets $D = \{(x_1, x_2, y)_{i=1}^N\}$ with a function of the form $y = f(x_1, x_2, \omega)$.



Multi-Node Perceptron

Given the complexity of ML models (i.e., ML functions), it is now common to use a graphical representation that uses higher-level components. For example,

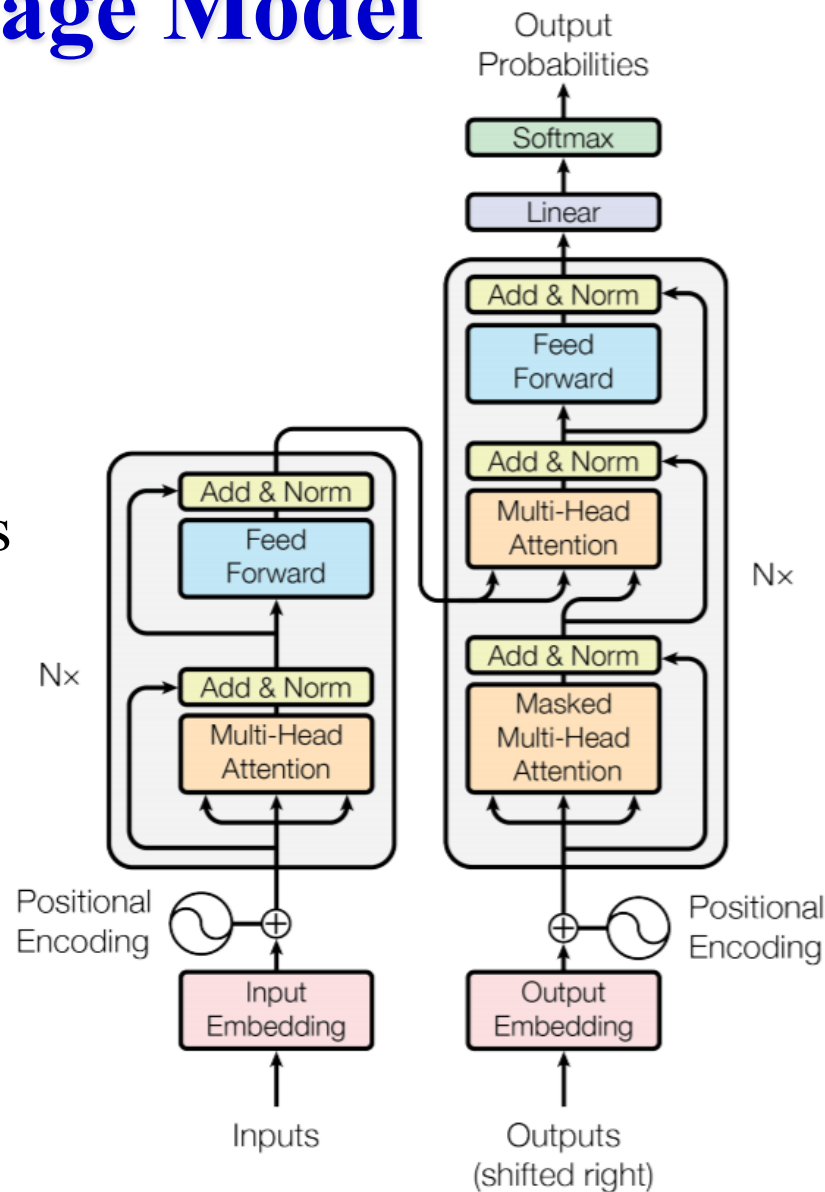


Large Language Model

Here, for example, is high-level graphical representation of an ML model called a **transformer**.

In ChatGPT 3, there are 96 layers with the same architecture, an instance of which is depicted as the two vertical grey boxes.

(Notice, the color-coding differs from ours!)



Summary

➤ General Approaches

- Supervised, unsupervised, generative, and reinforcement learning.

➤ Deep Learning

- Uses functions constructed through deep composition.

➤ The Perceptron

- Basic computational unit: matrix multiplication and addition followed by a nonlinear map.
- We described the multi-node perceptron and how machine learning models are represented graphically.